

A Predictive Analytics Study of Classification and Clustering Techniques for Patient Hospitalization Outcomes

Christina Loukidou

School of Medicine
Aristotle University
Thessaloniki, Greece
cloukida@auth.gr

1 INTRODUCTION

The objective of this study is to predict hospitalization outcomes and extract meaningful clinical information by developing methodologies in big data analytics. To reinforce the importance of health outcomes and understand patterns in patient data, we apply classification and clustering techniques. The analysis is based on six different research questions. Specifically, we evaluate the effectiveness of Decision Trees and Naive Bayes for forecasting clinical events and compare the performance of k-Nearest Neighbors (kNN) and Support Vector Machines (SVM) for mortality risk assessment. Furthermore, we implement k-Means, Hierarchical and DBSCAN algorithms, to segment patients and identify those who may require specialized care.

All data preprocessing, model implementation, data visualization and performance evaluations are conducted using R programming language. This process provides robust results to enhance clinical decision-making, patient triage and resource allocation in a hospital environment.

2 RESEARCH OVERVIEW

2.1 System Overview

The provided training and test datasets were first examined in order to verify variable types and ensure that categorical variables were appropriately converted as factors. The analysis was organized according to the six research questions. For each clinical question, only the useful variables were selected to take care that every model remained aligned with the experimental exploration.

2.2 Data Preprocessing

For the classification tasks, four different approaches were implemented. Decision Trees were used to predict the need for mechanical ventilation based on weight, SOFA and white blood cells measurements. Naïve Bayes classification was applied to predict sepsis occurrence using heart rate, systolic and diastolic blood pressure and liver health status. Also, two separate experiments were conducted in order to compare model performance with and without cardiovascular status. In addition, k-Nearest Neighbors and Support Vector Machines were used for mortality prediction based on age, weight, SOFA and glucose measurements. Since kNN and SVM are algorithms which compute distances, all variables must be in the same scale, so the numerical predictors were normalized before training. All classification models were evaluated using metrics, such as accuracy, precision, recall and F1 score.

For the clustering tasks, all numerical features were also scaled before every analysis. Firstly, k-means clustering was used to segment patients according to age, respiration rate and peripheral oxygen saturation, with the number of clusters being estimated through the SSE and silhouette methods. Hierarchical clustering was then performed on patients using SOFA score, weight, white blood cell counts and heart rate, using multiple linkage strategies, while Ward's method was maintained to interpret clusters structure. Finally, DBSCAN was used for density-based outlier detection identifying anomalous patient profiles based on glucose levels, white blood cell counts and SOFA score. Clustering analyses results were assessed through silhouette values, dendrogram interpretation and comparisons between cluster membership and important clinical outcomes such as sepsis mechanical ventilation and death.

3 EVALUATION

3.1 Evaluation Methodology

The first model, a Decision Tree Classifier, was developed, with the help of libraries `rpart` and `rpart.plot`, to predict the need for mechanical ventilation using weight, SOFA score and white blood cell measurements as predictors. The model was trained on the training dataset and tested on the test dataset. Three different `minsplit` values (10,20,30) were tested to examine the effect of model performance. The evaluation was assessed using classification metrics, including accuracy, precision, recall and F1 score. The decision tree was also interpreted offering important clinical information.

Naive Bayes classification was applied to predict sepsis occurrence using categorical clinical variables heart rate, systolic blood pressure, diastolic blood pressure and liver health status with use of libraries `e1071` and `MLmetrics`. Two models were evaluated: one excluding and one including cardiovascular status. Both models were trained on the training dataset and evaluated on the test dataset. Performance was assessed using accuracy, precision, recall and F1 score for both classes.

To address the third research question, k-Nearest Neighbors (kNN) and Support Vector Machines (SVM) were applied to predict patient mortality using age, weight, SOFA score and glucose levels. In this examination, libraries class, MLmetrics, ROCR and e1071 were used. Since both methods rely on distance calculations, all numeric features were standardized. For kNN, several values of k (1,2,3) were tested and k=1 was selected based on its performance in terms of recall and F1 score. For SVM, three kernel functions were evaluated: linear, radial (RBF) and polynomial. Model performance was assessed using precision, recall and F1 score for the positive class (death=YES). Additionally, ROC curves and AUC were used to compare models.

For the fourth research question, k-Means was applied to segment patients using age, respiration rates and oxygen saturation. All variables were again scaled for the analysis and library cluster was used. The optimal number of clusters was determined using the Elbow method for SSE and the maximum silhouette, indicating the best balance between cohesion and separation. Based on these criteria, a solution with k=2 clusters was selected. Clustering quality was evaluated using silhouette scores while the clinical characterization of clusters was assessed by examining their relationship with sepsis and mortality outcome.

Hierarchical clustering was applied with the help of library cluster, using weight, SOFA score, white blood cell counts and heart rate category. The categorical variable was transformed to numeric and all features were normalized for the analysis. Multiple linkage methods (single, complete, average and Ward's) were examined through dendrograms. Ward's method was selected due to its ability to produce more interpretable clusters. The optimal number of clusters was determined using silhouette method.

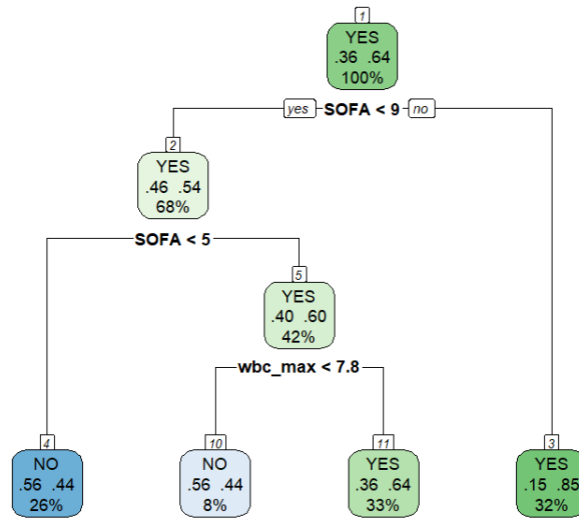
DBSCAN was applied to identify anomalous patient profiles using glucose levels, white blood cell counts and SOFA score with dbscan library. All features were standardized before the analysis. The eps parameter of the model was selected based on the k-Nearest Neighbors distance plot (eps=2) identifying the point where distances sharply increase and minPts was selected equal to the k used in the kNN plot securing the examination of the 10th closest neighbors. DBSCAN assigns patients to clusters with some of them representing outliers.

3.2 Evaluation Results

The results for the three different decision tree models using the minsplit values 10,20,30 showed same information, so one of them was retained to interpret. The model achieved accuracy of 0.67, indicating moderate predictive performance. As shown from Table 1, the model performs better for the positive class (mechanical ventilation = YES), with precision 0.743, recall 0.735 and F1 0.739. In contrast, performance for the negative class (mechanical ventilation = NO) is not that satisfying with precision 0.546 and recall 0.556. The decision tree (Figure 1) reveals that SOFA score is the most important predictor, with higher values strongly associated with the need for mechanical ventilation. Additional splits based on white blood cells counts further contribution to the prediction of outcome.

Table 1. Example Table

	Metrics for Decision Tree		
	Precision	Recall	F1
mech vent=NO	0.546	0.556	0.551
mech vent=YES	0.743	0.735	0.739

**Figure 2.** Decision Tree

The model without cardiovascular status achieved an accuracy of 0.754, but failed to identify any positive cases (sepsis=YES), resulting in recall equal to 0 and undefined precision in this class (Table 2). This indicates that the model predicts all instances as negative (sepsis=NO), likely due to class imbalance. When cardiovascular status was included, the accuracy remained similar (0.751), but the model was able to detect some positive cases, achieving precision 0.482 and recall 0.115 for this class (Table 3). However, performance for detecting remains low, while the model performs well for the negative class (recall 0.959).

Table 2. Confusion matrix for Naive Bayes model without cardiovascular status

Actual Class	Predicted Class	
	NO	YES
NO	1064	0
YES	348	0

Table 3. Naive Bayes model metrics with cardiovascular status

	Naive Bayes model metrics		
	Precision	Recall	F1
sepsis=NO	0.768	0.959	0.853
sepsis=YES	0.482	0.115	0.186

The kNN model with k=2 achieved the highest recall and F1 score among the tested values, indicating a better ability to identify patients at risk of death (Table 4). However, its overall accuracy was slightly lower compared to higher values of k, recall is more significant for this examination so k=2 model was retained. Among SVM models, the radial (RBF) kernel demonstrated the best overall performance, achieving higher precision (0.547), recall (0.17) and F1 score (0.26) compared to linear and polynomial kernels. ROC analysis showed that SVM (RBF) outperformed kNN, achieving a higher AUC (Figure 2). Depending on the clinical question we will use the appropriate model based on which metric is more important.

Table 4. kNN model metrics for different k values

k value	kNN model metrics		
	Precision	Recall	F1
1	0.356	0.376	0.365
2	0.373	0.392	0.382
3	0.409	0.362	0.362

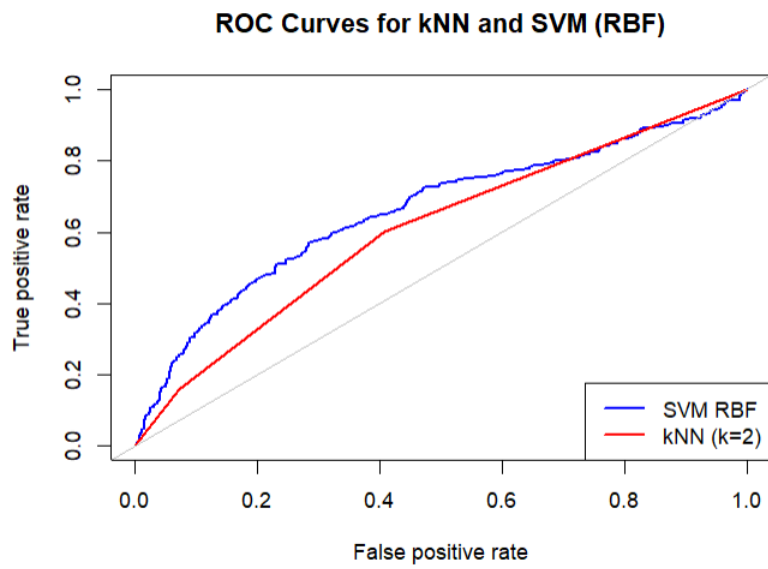


Figure 2. SVM (RBF) vs kNN

The SSE plot (Figure 3) shows a smoothly continuous reduction in SSE making it difficult to distinguish the best number of clusters, while the silhouette analysis achieves its highest value at $k=2$ (≈ 0.24) (Figure 4), indicating the most appropriate clustering solution. Cluster 1 is characterized by lower respiration rates and higher oxygen saturation, indicating a more stable respiratory condition. In contrast, cluster 2 presents higher respiration rates and lower oxygen levels, suggesting more severe clinical status. Also, second cluster exhibits higher sepsis (28.8%) and mortality (30.9%) rates, compared to first cluster (22.3% and 23.5%), indicating a patient group with higher risk (Tables 5,6).

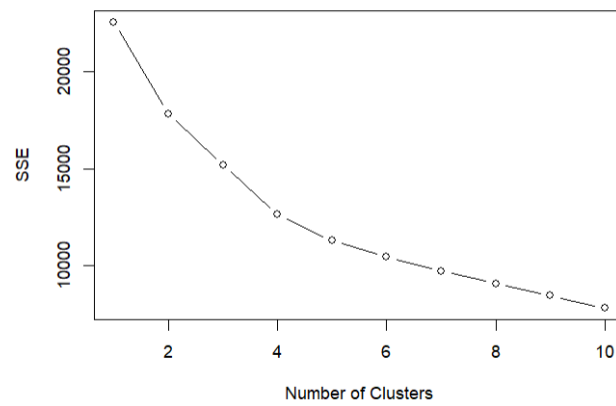


Figure 3. SSE plot

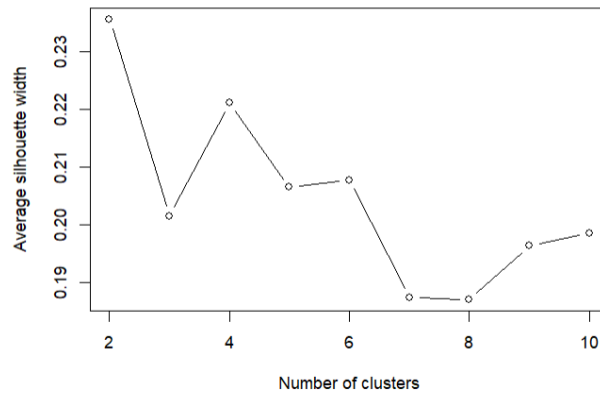


Figure 4. Silhouette analysis plot

Table 5. Percentages of two clusters for sepsis occurrence

Cluster	Sepsis occurrence	
	NO	YES
1	71.1%	28.8%
2	77.6%	22.3%

Table 6. Percentages of two clusters for mortality

Cluster	Mortality	
	NO	YES
1	69%	30.9%
2	76.4%	23.5%

The production of dendrograms for all methods showed that Ward's way of clustering is better from the others due to its interpretable clusters (Figures 5,6,7,8). The silhouette method for hierarchical analysis shows that $k=2$ provides the highest clustering quality (Figure 9), however this solution produces one dominant cluster of 5642 patients and a second very small cluster of 2 patients, suggesting the presence of outliers. Therefore, a $k=3$ was further examined for better interpretation. Cluster 1 is characterized by lower heart rate levels, lower white blood cell counts and slightly lower SOFA scores, indicating a relatively stable patient group. Cluster 2 includes patients with higher heart rate, higher white blood cell counts and slightly elevated SOFA scores. Cluster 3 contains a very small number of extreme cases with unusually high weight values, likely representing outliers (Table 7).

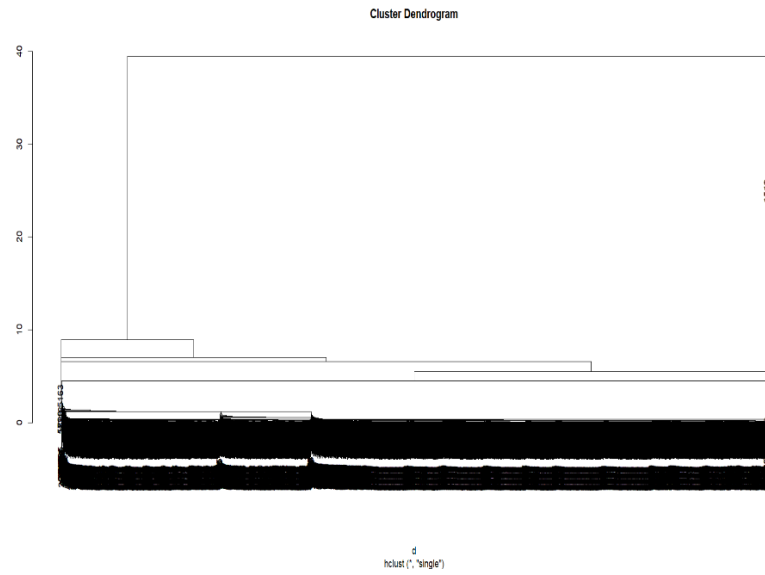


Figure 5. Dendrogram with single linkage

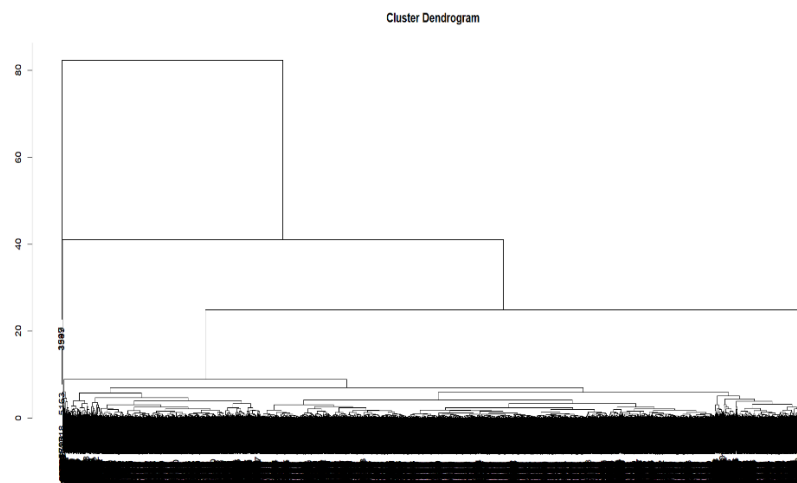


Figure 6. Dendrogram with complete linkage

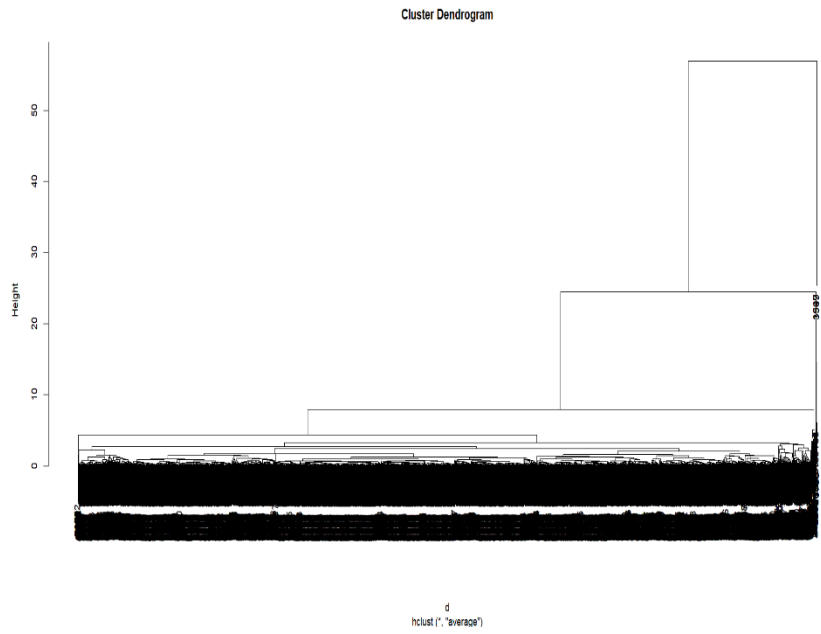


Figure 7. Dendrogram with average linkage

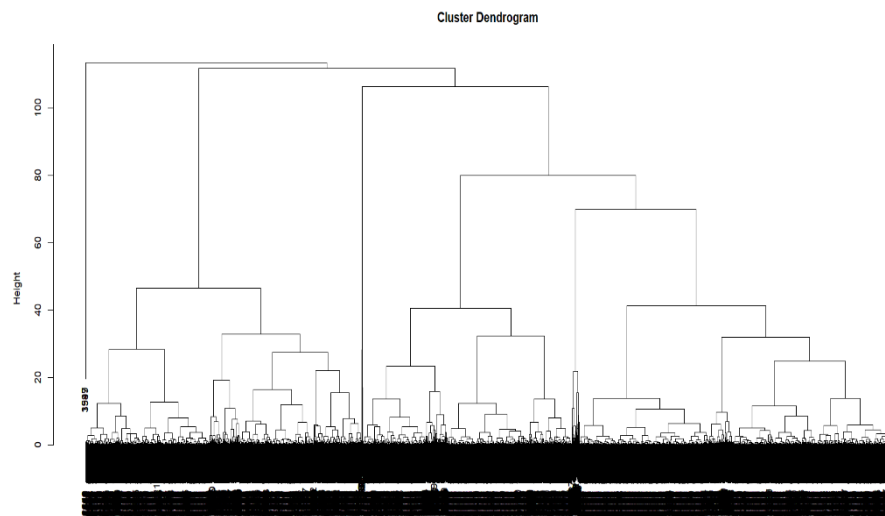


Figure 8. Dendrogram with Ward's method

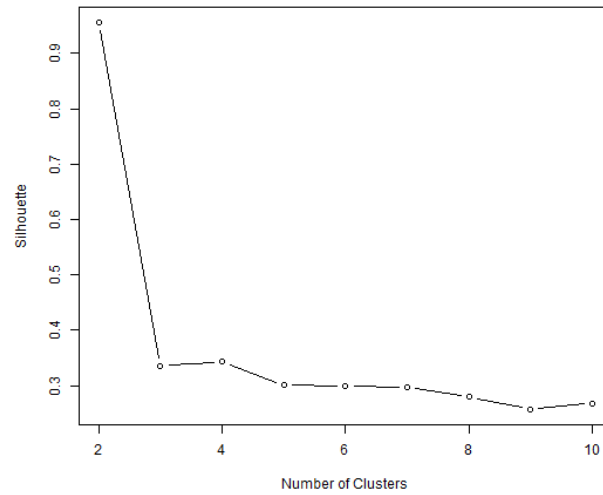


Figure 9. Silhouette analysis plot

Table 7. Rounded mean values of hierarchical clustering

Cluster	Mean values of features				
	weight	SOFA	wbc min	wbc max	hr num
1	82	7	9	11	1
2	82	7	13	17	3
3	11069	8	11	11	2

The kNN distance plot shows a clear elbow, supporting the chosen eps value(eps=2) (Figure 10). The model identified one main cluster of 5603 patients and one with 41 outliers. Outliers exhibit significantly higher values across all clinical variables compared to the overall population. Specifically, they show elevated glucose levels (glu_max $\approx$ 498 vs 193), white blood cell counts (wbc_max $\approx$ 120 vs 15.7) and SOFA score ($\approx$ 9.39 vs 7.26), indicating severe clinical differences. Finally, outliers are associated with higher rates of mechanical ventilation and mortality (Tables 8,9), indicating that anomalous patient profiles correspond to increased clinical risk.

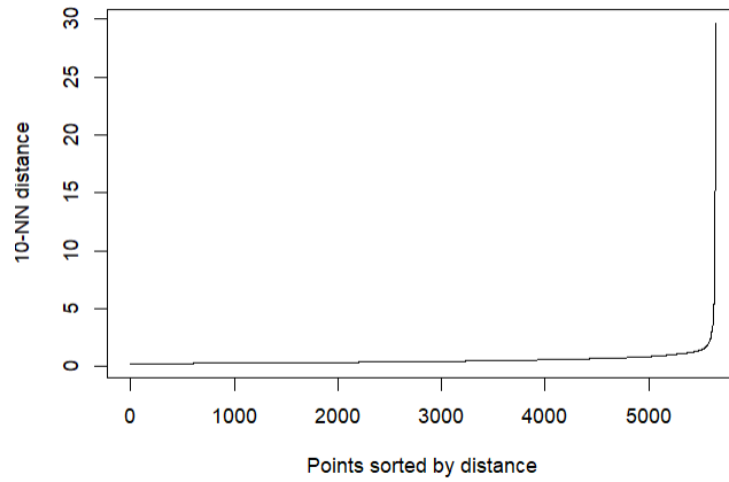


Figure 10. 10th kNN distance plot

Table 8. Percentages of two clusters for mechanical ventilation

Cluster	Mechanical ventilation	
	NO	YES
Outliers	36%	64%
Dominant cluster	26%	74%

Table 9. Percentages of two clusters for mortality

Cluster	Mortality	
	NO	YES
Outliers	74%	26%
Dominant cluster	63%	37%

4 CONCLUSIONS

The first model performs a moderate accuracy, but identifies very well patients who actually require mechanical ventilation, as reflected in the higher precision and recall for the positive class. This is clinically important, as higher recall for this class reduces the risk of missing cases who need instant care. At the same time, the lower

performance for the negative class indicates that the model may still produce misclassifications for low-risk patients. The decision tree highlights the significance of the SOFA score as the main predictor, followed by the maximum number of white blood cell measurements.

The Naive Bayes model shows limited effectiveness for predicting sepsis, particularly for the positive class. Although including cardiovascular status slightly improves detection of sepsis cases, the model still struggles to identify high-risk patients. This limitation is critical in a clinical context, where missing sepsis cases can have serious consequences. The results suggest that the independence assumption is not fully valid, as clinical variables such as blood pressure and cardiovascular condition are likely correlated.

Overall, kNN demonstrated competitive performance in mortality prediction and was preferred due to its higher recall and F1-score, making it more effective in identifying high-risk patients. Although SVM with a radial kernel achieved strong overall performance and provided a more robust model, kNN with $k=2$ offered a better balance for detecting positive cases. Therefore, kNN is considered more suitable for this task, while SVM remains a strong alternative for more stable and generalizable predictions.

The clustering analysis with k-Means successfully identified two clinically meaningful patient groups. Cluster 2, characterized by higher respiration rate and lower oxygen levels, demonstrated higher sepsis incidence and higher mortality. These findings suggest that k-Means clustering can support patient stratification and early risk identification. However, the moderate silhouette score shows that cluster separation is not strong and using additional clinical features may improve clustering performance.

Hierarchical clustering reveals a clear separation between lower-risk and higher-risk patient groups, primarily driven by heart rate and white blood cell levels. The dendrogram structure highlights how patients progressively merge into broader groups, which can support clinical decision-making. In practice, such hierarchical relationships can assist in patient triage, enable clinicians to identify high-risk individuals earlier and allocate medical resources more effectively.

DBSCAN effectively identifies a small group of highly abnormal patients with significantly worse clinical outcomes. These patients represent high-risk cases that may require immediate medical attention. From a clinical perspective, detecting outliers is crucial for early intervention, patient triage and efficient resource allocation.

5 SUGGESTIONS

Future work could explore more clinical feature additions and dimensionality reduction techniques to improve model performance, along with cross validation for more robust results. Furthermore, addressing class imbalance would enhance both predictive accuracy and clustering quality producing more reliable and significant results.

References

1. Springer Conference Proceedings Template, online: <https://www.overleaf.com/latex/templates/springer-conference-proceedings-template-updated-2022-01-12/wcvbtmwtjkj>